Analysis 1: Aug-Nov 2026

Prove the assertions from first principles.
Do not use any standard results without providing proof.

Exercises

(I) Let $f, g : [0,1] \mapsto [0,1]$ be differentiable functions and let f' and g' denote the derivatives. In each of the following cases show that h is also a differentiable function and express its derivative h' in terms of f' and g':

(a) $h(x) = \frac{1}{2}(f(x) + g(x), \;\; x \in [0,1]$.

(b) $h(x) = f(x)g(x), \;\; x \in [0,1]$.

(c) Suppose $g(x) \neq 0$ forall $x \in [0,1]$ and let $h(x) = \frac{f(x)}{g(x)}, \;\; x \in [0,1]$

(d) $h(x) = g(f(x)), \;\; x \in [0,1]$.

(II) Let $f : [a,b] \mapsto \mathbb{R}$ be a differentiable function and suppose $\exists c \in (a,b)$ such that $f(c) > f(a)$ and $f(c) > f(b)$. Show that $\exists u \in (a,b)$ such that $f'(u) = 0$.

(III) Let $f : [a,b] \mapsto \mathbb{R}$ be a differentiable function such that $f(a) = f(b)$. Show that $\exists u \in (a,b)$ such that $f'(u) = 0$.

(IV) Let $f : [a,b] \mapsto \mathbb{R}$ be a differentiable function. Show that $\exists u \in (a,b)$ such that $f'(u) = \frac{f(b)-f(a)}{(b-a)}$.

(V) Let $f(x) = x^3 + 5x^2 + 8$ and $g(x) = x^2 + 22x + 1$ for $x \in [0, \infty)$. Let $p(x) = g(f(x))$. Let $\epsilon = 0.01$. Let $x^* = 10$. Get $\delta > 0$ such that

$$0 < |h| < \delta \;\; \text{implies} \left| \frac{p(x^* + h) - p(x^*)}{h} - p'(x^*) \right| < \epsilon.$$

Here p' denotes the derivative of p.